

Ananmay Som Singh

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EDUCATION

University of Maryland, College Park | College Park, MD

Expected May 2028

Bachelor of Science in Computer Engineering

Relevant Coursework: Algorithms, Data Structures, Computer Systems, Compilers, Computer Organization, Linear Algebra, Probability & Statistics

EXPERIENCE

GSAlpha Labs | Real Estate Transaction Automation

May 2026 – Present

Software Engineering Intern · Part-time during academic year

San Francisco, CA

- Built HomeFlow AI, a Next.js and TypeScript platform that does the transaction-coordinator work California brokerages outsource at \$400–600 per file.
- Cut purchase-agreement intake from about 35 minutes of typing to under 30 seconds with an LLM pipeline that validates all 52 fields against a Zod schema. It scored 96% field-level accuracy on 15 real closed California transactions, and a human still signs off before filing.
- Scored OpenAI, Anthropic, and DeepSeek field by field on accuracy, latency, and cost per contract, then presented the tradeoff to the team. We shipped the model that held accuracy at 8x lower cost.
- Interviewed working California agents throughout the build. Their distrust of auto-filled deadlines is why every date now shows the rule that produced it.
- Automated contingency-deadline tracking across California business days and statutory holidays, where a missed date forfeits the buyer's earnest deposit. Logged every change with actor, field, and origin so disputes settle from the record; held 70% test coverage.

SEDS at UMD | SatFab CubeSat GPS Team

Sep 2024 – Feb 2026

Software Engineer

College Park, MD

- Built and documented the Python test framework for 26 Verilog modules in a CubeSat GPS receiver, trained 6 engineers on it, and it became the team's standard for every new module.
- Cut the full hardware regression from 4 hours to 95 minutes by running independent testbenches in parallel on a shared Linux server. What used to run overnight now runs before every merge.
- Automated waveform checking and pass/fail reports so integration bugs surfaced before anything reached the FPGA.

theconviction.ai | Financial Research Automation

May 2025 – Aug 2025

Software Engineering Intern

Remote

- Replaced 12 hours a week of analyst hand-collection with a Dockerized FastAPI and PostgreSQL pipeline that pulls SEC filings, earnings transcripts, and news for 50+ companies into one queryable record.
- Kept the pipeline running when sources failed, with timeouts, retries, and duplicate detection, and settled conflicts between overlapping feeds so each company had one clean record.
- Shipped a Next.js research tool the 4-person research team used daily. It puts each company on a timeline and links every finding to its source filing.

PROJECTS

Tape: Market Data Capture & Replay | Go, AWS S3, ECS, CloudWatch, Terraform, Docker

AnanmayS/tape

- Built a Go service that records live exchange feeds to S3 and replays them byte-identical at 2,580x real time, so a backtest on the same window always gives the same result.
- Detected all 3 gaps (3,240 messages lost) in fault-injection tests that severed the live feed every 25s, and flagged those windows so no backtest silently runs on missing data.
- Cut stored feed size 5.2x with a delta-encoded columnar format.
- Compared three backpressure strategies under load before picking one. It sustains 49.8k messages/sec against a 37/s live feed, roughly 1,000x headroom for market bursts.

ForgeGrid: Distributed Build System | Node.js, JavaScript, Docker, Linux

AnanmayS/forgegrid

- Designed a build system in Node.js that spreads a build across worker machines and starts each task the moment its dependencies finish.
- Measured 1 to 8 workers. Build time dropped 59% but stopped improving past 3, since the longest chain of dependent tasks sets the floor.
- Added a content-addressed cache that skips any task whose inputs have not changed, so a repeat build only redoes what actually differs.
- Survived worker crashes by reassigning their tasks, proven by tests that kill workers mid-build and check the output is still right.

ShowdownRL: Reinforcement Learning Agent | Python, PyTorch, PPO, Playwright, Gymnasium

AnanmayS/ShowdownRL

- Trained a PPO agent on a 106-feature game view that won 79% of 1,000 live matches against humans, and masked illegal moves so it never wastes a turn.
- Drove real matches through the browser with Playwright and saved every battle log, so any reported win rate traces back to the games behind it.

TECHNICAL SKILLS

Languages: Go, Python, Java, C++, C, TypeScript, JavaScript, SQL, Verilog

Backend & Data: FastAPI, Node.js, PostgreSQL, Drizzle, SQLAlchemy, Supabase, Zod, WebSockets, Concurrency

Infrastructure & Tooling: AWS, S3, ECS, CloudWatch, Terraform, Docker, Linux, GitHub Actions (CI/CD), Vercel, pytest, Vitest

Frameworks & ML: React, Next.js, PyTorch, LLM extraction and evaluation